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HUNNY GUPTA
9999903256, 9999603256

HT Board Booster

Class X Science

Biology

CBSE Sample Paper Question Bank

Chapter wise Practice

Chapters 5 - 6
Life Process and Control and Coordination

Session 2026-27

Practice | Perform | Progress

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Hunny Gupta

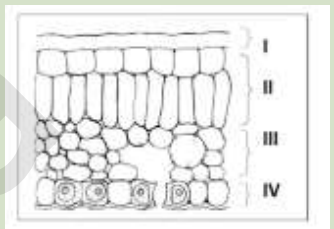
Class X Biology
CBSE SQP 2019-2025
Ch- Life Process

SECTION – A

Q.1. Veins are thin walled and have valves. Justify. **(CBSE SQP 2020-21)**

Q.2. State the role of pancreas in digestion of food. **(CBSE SQP 2020-21)** (1)

Q.3. In the given transverse section of the leaf identify the layer of cells, where maximum photosynthesis occurs. **(CBSE SQP 2022-23)**

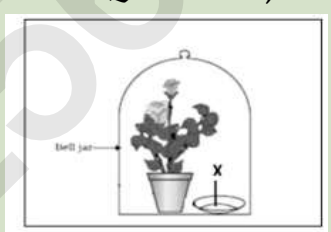


(a) I, II (b) II, III (c) III, IV (d) I, IV

Q.4. **Assertion:** Amphibians can tolerate mixing of oxygenated and deoxygenated blood. **(CBSE SQP 2022-23)**

Reason: Amphibians are animals with two chambered heart

Q.5. Observe the experimental setup shown below. Name the chemical indicated as 'X' that can absorb the gas which is evolved as a by product of respiration. **(CBSE SQP 2022-23)**



(a) NaOH (b) KOH (c) $Ca(OH)_2$ (d) K_2CO_3

Q.6. Generally food is broken and absorbed within the body of organisms. In which of the following organisms is it done outside the body? **(CBSE SQP 2023-24)**

(a) Amoeba (b) Mushroom (c) Paramecium (d) Lice

Q.7. A sportsman, after a long break of his routine exercise, suffered muscular cramps during a heavy exercise session. This happened due to: **(CBSE SQP 2023-24)**

a) lack of carbon dioxide and formation of pyruvate.
b) presence of oxygen and formation of ethanol.
c) lack of oxygen and formation of lactic acid.
d) lack of oxygen and formation of carbon dioxide

Q.8. During cellular oxidation of Glucose, ATP is produced along with formation of other products in this reaction. Which of the following events is associated with production of maximum ATP molecules per molecule of Glucose during this process? Synthesis of _____. **(CBSE SQP 2024-25)**

a) ethanol in yeast b) lactic acid in muscle cells
c) carbon dioxide in yeast cells d) carbon dioxide in human cells

Q.9. During which of the following stages of the circulation of blood in a normal human being, the oxygenated blood is pumped to all parts of the body? **(CBSE SQP 2024-25)**

a) contraction of the left atrium b) contraction of left ventricle
c) relaxation of the right atrium d) relaxation of the right ventricle

Q.10. Which of the following adaptations in herbivores helps in digestions of cellulose? **(CBSE SQP 2024-25)**

a) Longer large intestine b) Smaller large intestine
c) Smaller small intestine d) Longer small intestine

Q.11. Select the group in which all organisms have the same mode of nutrition. **(CBSE SQP 2025-26)**

a) Cuscuta, yeast, legumes, leeches and tapeworm
b) Cactus, ticks, lice, leeches and cow
c) Cuscuta, ticks, lice, leeches and tapeworm
d) Cactus, grass, lice, lion and tapeworm

Q.12. Which of the following options indicates the products formed after breakdown of the glucose in our muscle cells when there is lack of oxygen? **(CBSE SQP 2025-26)**

a) Ethanol + carbon dioxide + Energy b) Lactic acid + Energy
c) Lactic acid + carbon monoxide + Energy d) Carbon dioxide + Water + Energy

SECTION - B

Q.13. Bile juice does not have any digestive enzyme but still plays a significant role in the process of digestion. Justify the statement. **(CBSE SQP 2020-21)**

OR

In birds and mammals the left and right side of the heart are separated. Give reasons.

Q.14. Patients whose gall bladder is removed are recommended to eat less oily food. Why? **(CBSE SQP 2022-23)**

Q.15. Name the substances other than water that are reabsorbed during urine formation. What are the two parameters that decide the amount of water that is reabsorbed in the kidney? **(CBSE SQP 2022-23)**

Q.16. What is the purpose of making urine in the human body? Name the organs that stores and releases the urine. **(CBSE SQP 2023-24)**

OR

Why do arteries have thick and elastic walls whereas veins have valves?

Q.17. Differentiate between alveoli and nephron on the basis of the following points

S.No	Feature	Alveoli	Nephron
1	Structure and location		
2	Function		

(CBSE SQP 2024-25)

Q.18. List the steps for the synthesis of glucose by the plants. What special feature is found in desert plants related to this process? **(CBSE SQP 2024-25)**

OR

Explain the role of the following enzymes in the process of digestion of food in humans:
(i) Salivary amylase (ii) Pepsin (iii) Trypsin (iv) Lipase

Q.19. Unlike animals, plants do not have any excretory products as they do not eat food. Comment upon the statement with justification. **(CBSE SQP 2025-26)**

Q.20. How many chambers are there in the heart of the following organisms? How is mixing of oxygenated and deoxygenated blood prevented in their body? **(CBSE SQP 2025-26)**

(i) Fishes (ii) Humans

OR

Explain the mechanism by which the water is transported in plants?

SECTION - C

Q.21. Explain the processes of aerobic respiration in mitochondria of a cell and anaerobic respiration in yeast and muscle with the help of word equations. **(CBSE SQP 2019-20)**

Q.22. The leaves of a plant were covered with aluminum foil; how would it affect the physiology of the plant?

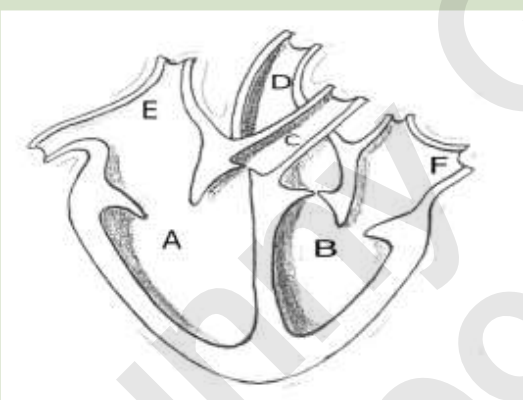
OR

How is lymph an important fluid involved in transportation? If lymphatic vessels get blocked, how would it affect the human body? Elaborate. **(CBSE SQP 2022-23)**

Q.23. Water is used by the leaves of the plants for photosynthesis but rather than watering the leaves, we water the plant through the soil. How does this water reach the leaves of the plant? **(CBSE SQP 2024-25)**

SECTION – D

Q.24.



(i) Identify any two parts from the above diagram which carry oxygenated and deoxygenated blood. **(CBSE SQP 2019-20)**

(ii) Explain the process of double circulation with the help of a flow chart.

Q.25. Neha consumed boiled sweet potatoes and boiled eggs for breakfast. Help her to understand some steps in the process of digestion of the food taken by her by answering the questions given below. **(CBSE SQP 2025-26)**

Attempt either subpart A or B.

A. Which of these food items is rich in proteins? In which part of the alimentary canal is the digestion of this component initiated? Name the enzymes, conditions required and the glands associated with the digestion here.

OR

B. Which of these food items contains fats? How is it digested?

C. Which of these food items is rich in starch? How is its digestion initiated?

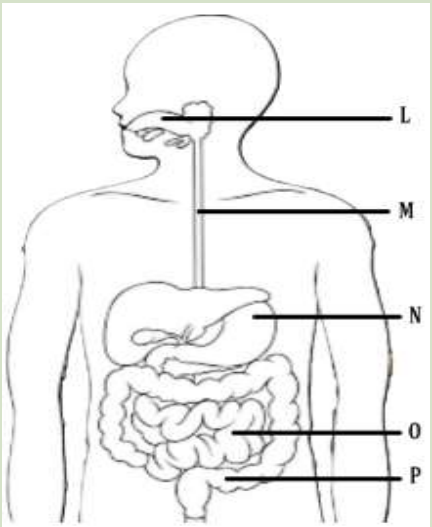


Figure: Human Alimentary canal

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D. The figure given below represents parts of the human alimentary canal. Which of these parts will have the maximum amount of digested food as soon as the process of digestion is completed?

SECTION – E

Q.26. Question numbers 4(a) - 4(d) are based on the two tables given below. Study these tables related to blood sugar levels and answer the questions that follow.

Table A (Blood glucose chart)

	Mean Blood Glucose Level (mg/dL)
Doctor's advice needed	380
	350
	315
	280
	250
	215
Good	180
	150
Excellent	115
	80
	50

Total B (Blood Report of Patient X and Y)

Time of check	**Blood Glucose ranges (mg/dL)**	
	Patient X	**Patient Y**
Before breakfast (Fasting)	<100	70-130
Before lunch, supper and snack	<100	70-130
Two hours after meals	<140	<180
Bedtime	<120	90-15

4(a) Refer to Table B showing the blood report of the levels of glucose of patients X and Y. Infer the disease which can be diagnosed from the given data. (1)

4(b) Identify the hormone whose level in the blood is responsible for the above disease. (1)

4(c) Which one of the following diets would you recommended to the affected patient?

i) High sugar and low fat diet. ii) Low sugar and high protein diet.
iii) High Fat and low fiber diet. iv Low sugar and high fiber diet. (1)

4(d) Refer to the Table A and suggest the value of the mean blood glucose level beyond which doctor's advice is necessary: (1)

i) 180 mg/dL ii) 115 mg/dL
iii) 50 mg/dL iv) 80 mg/dL

Q.27. Read the following and answer any **four** questions from 17 (i) to 17 (v)
All living cells require energy for various activities. This energy is available by the breakdown of simple carbohydrates either using oxygen or without using oxygen.

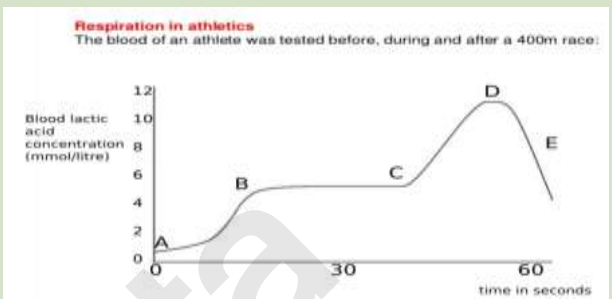
(i) Energy in the case of higher plants and animals is obtained by

a) Breathing b) Tissue respiration
c) Organ respiration d) Digestion of food

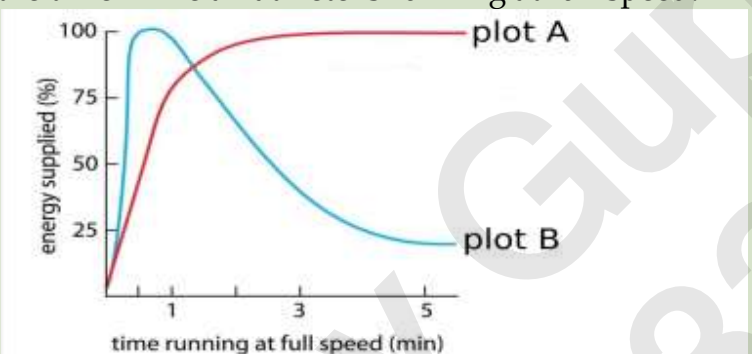
(ii) The graph below represents the blood lactic acid concentration of an athlete during a race of 400 m and shows a peak at point D.

Lactic acid production has occurred in the athlete while running in the 400 m race. Which of the following processes explains this event?

a) Aerobic respiration
b) Anaerobic respiration
c) Fermentation
d) Breathing



(iii) Study the graph below that represents the amount of energy supplied with respect to the time while an athlete is running at full speed.



Choose the correct combination of plots and justification provided in the following table.

	Plot A	Plot B	Justification
(a)	Aerobic	Anaerobic	Amount of energy is low and inconsistent in aerobic and high in anaerobic.
(b)	Aerobic	Anaerobic	Amount of energy is high and consistent in aerobic and low in anaerobic
(c)	Anaerobic	Aerobic	Amount of energy is high and consistent in aerobic and low in anaerobic
(d)	Anaerobic	Aerobic	Amount of energy is high and inconsistent in anaerobic and low in aerobic

(iv) The characteristic processes observed in anaerobic respiration are

(i) presence of oxygen (ii) release of carbon dioxide
(iii) release of energy (iv) release of lactic acid

(a) (i), (ii) only (b) (i), (ii), (iii) only (c) (ii), (iii), (iv) only (d) (iv) only

(v) Study the table below and select the row that has the incorrect information.

		Aerobic	Anaerobic
(a)	Location	Cytoplasm	Mitochondria
(b)	End Product	CO_2 and H_20	Ethanol and CO_2
(c)	Amount of ATP	High	Low
(d)	Oxygen	Needed	Not needed

Hunny Gupta 9999803256

Class X Biology
CBSE SQP 2019-2025
Ch.- 6 Control and Coordination

Section - A

Q.1. Observe the three figures given below. Which of the following depicts tropic movements appropriately? **(CBSE SQP 2022-23)**

(a) B and C
(b) A and C
(c) B only
(d) C only



Q.2. Receptors are usually located in sense organs. Gustatory receptors are present in
(a) Tongue (b) nose (c) eye (d) ear
(CBSE SQP 2023-24)

Q.3. Height of a plant is regulated by **(CBSE SQP 2023-24)**
(a) DNA which is directly influenced by growth hormone
(b) Genes which regulate the proteins directly.
(c) Growth hormones under the influence of the enzymes coded by a gene.
(d) Growth hormones directly under the influence a gene.

Q.4. There was a cerebellar dysfunction in a patient. Which of the following activities will get disturbed in this patient as a result of this? **(CBSE SQP 2024-25)**
(a) Salivation (b) Hunger control
(c) Posture and balance (d) Regulation of blood pressure

Q.5. Which of the following is a correct combination of function and part of the brain?
(a) Posture and balance: Cerebrum **(CBSE SQP 2025-26)**
(b) Salivation: Medulla in midbrain
(c) Hunger: Pons in hindbrain
(d) Blood pressure: Medulla in hindbrain

Q.6. The blood glucose level in a patient was very high. It may be due to inadequate secretion
(a) Growth hormone from pituitary gland **(CBSE SQP 2025-26)**
(b) oestrogen from ovary
(c) Insulin from pituitary gland
(d) Insulin from pancreas

Section - B

Q.7. How is the mode of action in beating of the heart different from reflex actions? Give four examples. **(CBSE SQP 2022-23)**

Section - C

Q.8. Explain giving reasons the bending of the shoot tip of a plant towards light source coming from one side of the plant. **(CBSE SQP 2019-20)**

Q.9. We are advised to take iodised salt in our diet by doctors. Justify its importance in our body **(CBSE SQP 2023-24)**

Q.10. Draw and explain how the nerve cells help in transmission of impulses? **(CBSE SQP 2025-26)**

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Section – D

Q.11. a) A doctor has advised Sameer to reduce sugar intake in his diet and do regular exercise after checking his blood test reports. Which disease do you think Sameer is suffering from? Name the hormone responsible for this disease and the organ producing the hormone.
b) Which hormone is present in the areas of rapid cell division in a plant and which hormone inhibits the growth? **(CBSE SQP 2023-24)**

Section – E

Q.12. Mohan and Rohit observed that shoots of a plant growing in shade bend towards the sunlight. Whereas, leaves of 'Touch me not' plant fold and droop soon after touching. They were curious to know how these movements occur in plants. **(CBSE SQP 2024-25)**



A. Shoots of a plant bending towards light



B. Folding of leaves Touch me not plant

In order to help them understand the movements in the plants, answer the following questions:

Attempt either subpart A or B.

A. What causes the bending of shoots in the plants as shown in figure A?

OR

B. What causes the folding of the leaves in 'Touch me not' plant as shown in figure B?

C. Compare the movement of growth of the pollen tube towards ovule with he movements shown in part A of the above figure

D. Compare the movement shown in figure B with the movement of body parts in the animals.

Hunny Gupta 9999803256